

Swiss Water Monitoring Networks

Rivers & Streams

National and cantonal networks monitoring rivers, streams, and surface watercourses

20 networks in this file	6 Sept 2026 document date	Research: 31 Aug 2026 data research date
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This document is part of the Swiss Water Monitoring Networks research series (41 networks mapped, research date: 31 August 2026). It covers 20 river/stream monitoring networks. Every network entry was sourced from fetched agency web pages; source URLs are embedded in the table cells of the full research document. Generated by Perplexity Computer.

Rivers & Streams — 20 Networks

#	Agency	Network	Water Body Type	Parameters	Stations	Frequency	Format	Open Data	API / Access
1	BAFU/FOEN	Basismessnetz Oberflächengewässer	Rivers + lakes	Water level; discharge; water temperature; chemical composition; turbidity	~260 stations (200 discharge sites)	Continuous; 10-min updates via LINDAS	XLSX (station dirs); ESRI SHP/FileGDB	Yes — Opendata OPEN: Freie Nutzung	STAC/LINDAS; hydro daten.admin.ch; map.geo.admin.ch
2	BAFU/FOEN	Messnetz Wassertemperatur (Fließgewässer)	Rivers	Water temperature	~80 stations	Continuous (kontinuierlich)	API (JSON) / XML via opendata.swiss	Partial — published on opendata.swiss; no licence named	hydrodaten.admin.ch ; API (JSON)/XML; republished via api.existenz.ch
4	BAFU/FOEN	Hydrologische Untersuchungsgebiete (HUG)	Rivers in near-natural catchments	Natural runoff depths; catchment precipitation; evaporation	~40 study catchments	Monthly/annual aggregates (since 1957)	n.a.	Station dataset on opendata.swiss	opendata.swiss; station geodata on geo.admin.ch
5	BAFU/FOEN + cantons	NAWA TREND	Rivers and streams	Nutrients; micropollutants; biological indicators; chem-phys parameters	~140 monitoring sites	Long-term continuous (chem-phys since 2011; biology since 2012)	Station list as XLSX	Partial — station lists published; no measurement-data licence	BAFU NAWA page downloads; no API
6	BAFU + Eawag + WSL	NAWA FRACHT (formerly NADUF)	Rivers (large, medium, smaller watercourses)	Water level; discharge; water temperature; pH; conductivity; DO; 14-day composites: hardness, nutrients, DOC, TOC, heavy metals	Selected BAFU stations + 3 WSL Alptal (Erlenbach, Vogelbach, Lümpenbach)	Sensors continuous; chemistry 14-day composites; Alptal weekly	Programme docs as PDF; no measurement-data download stated	Partial — programme docs published; data licence not stated	BAFU NAWA page; no API
7	BAFU/FOEN	NAWA SPEZ	Rivers, incl. small streams	Pesticides; biological state of small streams	Selected sites (number not stated)	Campaign-based (2012, 2015, 2017, 2023 pesticides; 2018 biology)	n.a.	n.a.	n.a.
12	BAFU/FOEN (portal)	hydrodaten.admin.ch — Hydrologische Daten	Rivers, lakes, groundwater (portal)	Water level, discharge, water temperature, flood danger, water quality	Aggregates BAFU networks; no own station count	Real-time raw data; LINDAS updated every 10 minutes	API (JSON)/XML on opendata.swiss; LINDAS Linked Data Service	Partial — published as open data via LINDAS; no licence name on portal	LINDAS; 'Meine Pegel' app; station web pages
15	AUE Basel-Stadt + BAFU + LUBW	Rheinüberwachungsstation Weil am Rhein (RUS)	River (Rhine, km 171.37)	680 pollutants (420 daily); continuous sensor parameters (temp, pH, conductivity, etc.)	1 binational station	Continuous; averaged every 15 min; annually plausibilised hourly values	Open-data dataset 100046 with API (data.bs.ch)	Yes — publicly accessible via Basel-Stadt open-data portal	data.bs.ch dataset 100046 + API console; Birs dataset 100236
16	AWEL Kanton Zürich (Hydrometrie)	Hydrometrische Stationen (kantonaless Messnetz)	Rivers + lakes	Water level; discharge (from rating curve)	Over 60 stations	Continuous near real-time; curves last 24 h / 7 d / 31 d	Delivery on request; metadata at maps.zh.ch	Partial — declared OGD; delivery via order form; no API	Web viewer + order form; maps.zh.ch; attribution 'AWEL, Kanton Zürich'

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17	AWEL Kanton Zürich (Gewässerschutzlabor)	Gewässermonitoring / Untersuchungsprogramm Seen u. Fließgewässer 2022–25	Lakes, rivers, groundwater + springs	Water quality (Modul-Stufen-Konzept); micropollutants; ecomorphology	11 lakes; 150 river sites; ~100 groundwater/spring abstractions	Continuous auto-sampling at 15 main stations; periodic at 150 others	PDF reports	No — reports/maps only; no API/licence stated	Web maps with per-station reports
20	AWA Kanton Bern	Hydrologische Messnetze des AWA	Rivers, lakes, groundwater, precipitation	Discharge; water level; water temperature; groundwater level/temperature; precipitation	~50 discharge; 90 groundwater; 120 level gauges; 15 precipitation	n.a.; monthly groundwater bulletin	Station lists via cantonal geoportal; bulletins and yearbooks	Partial — data published online with federal data; delivery via Datenbezug	Geoportal maps + order channel; flood portal
21	DGE Canton de Vaud	Suivi de la qualité des rivières	Rivers	Aquatic macroinvertebrates (biological quality); nutrients (C, N, P); micropollutants (pesticides, pharmaceuticals)	204 sites on >50 rivers	n.a. (since 1990s)	Web platform; reports as PDF	Partial — results on vhv-qualite.ch; no licence/API stated	vhv-qualite.ch; Géoportail de la veille hydrologique vaudoise
22	Canton Ticino, UCA	Rete di misura cantonale (idrometria)	Rivers + lakes (Lago Maggiore, Lago di Lugano)	Discharge (m³/s); water temperature; lake level; precipitation	24 hydrometric + 31 meteorological cantonal stations; OASI lists 39 river stations	Half-hourly, daily, monthly, annual values; quarterly bulletin; annual yearbook	Portal download (OASI); yearbooks as PDF	Partial — free portal download; no licence/API	OASI web portal (station + parameter + interval selection + download)
23	Canton Ticino, UPAAI	Monitoraggio corsi d'acqua — Sistema modulare a livelli	Rivers/streams	Nitrogen compounds; total phosphorus; external aspects module; ecomorphology	986 watercourses analysed (17% of ~5,800 km cantonal network)	Campaign-based (2003–2005; 2021–2022)	Field check-lists; no data formats stated	No — no licence /API/download stated	n.a.
24	Canton de Genève	Veille hydrologique / portail hydro-météorologique	Lake + rivers	Precipitation; water level; discharge; lake level; temperature; low-flow indicators	37 stations: 12 rain-gauge + 25 limnimetric	Real-time; monthly bulletins; weekly low-flow follow-up in summer	n.a.	Partial — historical data automatic extraction; no licence stated	Web portal with automatic historical extraction; station geodata on SITG
25	Kanton Aargau (Hydrometrie)	Hydrometrisches Messnetz Aargau	Rivers	Water level (redundant: pressure + radar); water temperature	n.a. — multiple stations with individual pages	Transmitted every 10 minutes; time-series download at 10-min resolution	Standardised formats (names not stated)	Yes — freier Gebrauch; no licence name stated	Station web app ag.ch/app/hydrometrie/station/
27	Kanton Luzern (uwe + vif)	Wassermessstellen / Hydrometrie Luzern	Rivers, lakes, groundwater, springs (+ precipitation)	River water level and discharge; lake level; groundwater level; spring discharge; precipitation	14 river sites; ~40 groundwater/spring sites; 10 precipitation stations	n.a.; records partly back to 1970	Station geodata: ESRI GDB, SHP, OGC GeoPackage; other formats chargeable	Partial — station geodata on geoportal; measurement data via order form (chargeable)	geo.lu.ch geoportal + Bestellung Hydrologiedaten form

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2 8	Kanton Thurgau, Amt für Umwelt	Hydrologisches Messnetz Thurgau	Rivers/groundwater	Water level (m a.s.l.); water temperature	n.a. — various stations	Live datasets: last 365 days; finer granularity in full downloads	Zipped CSV; JSON schema documented	Yes — published on cantonal OGD portal (mirrored on opendata.swiss); no licence name	Opendatasoft portal + API console; ThurGIS station map
3 6	WSL	Hydrometeorological long-term monitoring, Alptal (Erlenbach, Vogelbach, Lümpenenbach)	Mountain torrents + groundwater	Hydrology (runoff); meteorology; snow; water quality; bedload; groundwater; isotopes; nutrients	3 torrent catchments; snow at >15 locations	10-minute means since 1984; daily means 1968–2017; snow weekly/monthly; water quality weekly	EnviDat download (daily means); sub-daily on request	Partial — daily means openly downloadable via EnviDat; sub-daily on request	envidat.ch downloads + researcher requests
3 8	Existenz.ch (community project)	BAFU Hydrology API + SwissMetNet API + custom sensors	Rivers (surface water) + weather	Surface-water temperature; water flow; water level; air temperature; humidity; precipitation; wind; pressure	'Many measurement stations'; exact count not stated	10-minute source updates; polling every 5 min recommended	JSON; CSV via Datasette; InfluxDB 2 archive (Flux queries); RSS	Partial — free for public/non-commercial; BAFU data must be credited+linked	REST JSON endpoints (CORS *); Datasette; InfluxDB archive; open-source parsers

Agency Notes

BAFU / FOEN (Bundesamt für Umwelt)

FOEN is the backbone of Swiss water monitoring, operating and coordinating several networks combining long-term fixed stations with temporary sites. The quantity side (Basismessnetz, ~260 stations, 200 discharge sites) is 90% automated with remote retrieval and calibrated by 5–6 discharge measurements per station per year plus weekly on-site checks. Station geodata are published as 'Opendata OPEN: Freie Nutzung' via geocat.ch; real-time values via LINDAS at 10-minute cadence. The SMS alerting service ended 1 July 2026 — users directed to 'Meine Pegel' push notifications. NADUF was renamed NAWA FRACHT in 2025. Contact: hydrologie@bafu.admin.ch / abfragezentrale@bafu.admin.ch

NAWA — National Surface Water Quality Monitoring Programme

NAWA is explicitly limited to flowing waters and does not replace cantonal surveys. It comprises three modules: TREND (~140 sites, uniform parameter set, chem-phys since 2011, biology since 2012, micropollutants since 2018); FRACHT/NADUF (continuous electrode sensors plus 14-day composite chemistry, run jointly with Eawag and WSL since 1972/2003); and SPEZ (campaign monitoring: pesticides 2012/2015/2017/2023, small-stream biology 2018). FOEN maintains a published parameter-name list for chemical substances (version 11/2025). Source: bafu.admin.ch/nawa

AUE Basel-Stadt / Rheinüberwachungsstation Weil am Rhein (RÜS)

Operated by AUE Basel-Stadt as part of the international Rhine warning and alarm plan (IWAP) with the International Main Warning Centre. Binational (CH/DE) station at Rhine km 171.37 monitoring 680 pollutants (420 daily). Data published on data.bs.ch as dataset 100046 at 15-minute raw resolution with annually plausibilised hourly values. Additional Basel-Stadt datasets: Birs (100236), Rhine level (100089).

AWEL Kanton Zürich (Amt für Abfall, Wasser, Energie und Luft)

Four distinct activities: (1) Hydrometry: >60 level stations, near-real-time, discharge derived from rating curves — declared OGD but delivered via order form (free for normal requests, CHF 140.20/h for very large); no API, attribution 'AWEL, Kanton Zürich' required. (2) Surface-water quality: 15 continuously auto-sampled main stations + 150 periodic sites + 11 lakes + ~100 groundwater/spring abstractions on 4-year rotation. (3) Groundwater levels: continuous data loggers. (4) Groundwater quality: nitrate, conductivity, temperature, O₂, pH, ions, PFAS, chlorothalonil metabolites since 2019. ISO 17025 accredited. Contact: marco.walser@bd.zh.ch

AWA Kanton Bern (Amt für Wasser und Abfall)

Best-quantified cantonal network: ~50 discharge stations, 90 groundwater stations, 120 level gauges and 15 precipitation stations. Publishes hydrographic bulletins/yearbooks and monthly regional groundwater evaluation. Data published online together with federal data; series obtained through 'Datenbezug' channel. Federal NAQUA sites in Bern: Niederbipp, Utzenstorf, Worben. Source: bvd.be.ch

DGE Canton de Vaud (Direction générale de l'environnement)

River-quality programme covers 204 sites on >50 rivers, since the 1990s, with biological (macroinvertebrate) and chemical (nutrients, micropollutants) assessment. Since 2024 results are public on vhv-qualite.ch, complementing the 'Géoportail de la veille hydrologique vaudoise' for quantity data. Terminology note: 'SESA' (Service des eaux, sols et assainissement) is a Vaud service. vhv-qualite.ch timed out when fetched on 31.08.2026.

Canton Ticino — UCA and UPAAI

Two separate agencies: Ufficio dei corsi d'acqua (UCA) runs the quantity network (24 hydrometric + 31 meteorological stations, integrated with FOEN and MeteoSwiss; OASI portal exposes 39 river stations with half-hourly to annual data and download). Ufficio protezione acque e approvvigionamento idrico (UPAAI) runs quality/ecomorphology using the federal Modular Stepwise Procedure: 986 watercourses (17% of ~5,800 km network) analysed, campaigns 2003–2005 and 2021–2022. Critical nitrogen levels found downstream of WWTP outfalls in Vedeggio, Lavaggio and Breggia.

Canton de Genève — Veille hydrologique

37 stations (12 rain gauges + 25 limnimetric), real-time portal with automatic historical extraction per station, annual validation lag. Portal integrates MeteoSwiss, FOEN and SIG measurements. Station geodata on SITG. Source: ge.ch

Kanton Aargau — Hydrometrie and Grundwasser

Most transparent sensor documentation of any Swiss canton: redundant pressure + radar level probes, water-temperature sensing, PLC cabinets with dual mobile/leased-line transmission, 10-minute data free for public use. Groundwater: >50 online abstraction stations at daily resolution, joint precipitation network with Luzern. Some AG stations are FOEN Basismessnetz sites. Source: ag.ch

Kanton Luzern — Hydrometrie

14 river sites, ~40 groundwater/spring sites, 10 cantonal precipitation stations; records back to 1970. Station geodata in ESRI GDB/SHP/GPKG. Data delivery is form-based and chargeable. Canton owns the data except federal/MeteoSwiss sites; users warned to verify series for design purposes. Source: uwe.lu.ch

Kanton Thurgau — Hydrologisches Messnetz

Most API-friendly canton: hydrometry stations, water level and temperature as separate OGD datasets on data.tg.ch with zipped-CSV bulk download and JSON schema, joined via obj_id. Published values are unverified raw sensor data, not plausibility-checked. Network shared reporting with Schaffhausen. Source: data.tg.ch

WSL (Swiss Federal Institute for Forest, Snow and Landscape Research)

Water sensing concentrated in the Alptal torrent research site (Erlenbach, Vogelbach, Lümpenenbach): 10-minute means since 1984, daily means 1968–2017 downloadable from EnviDat, weekly NADUF water quality since 2003, snow at >15 manual locations, plus bedload, isotope (ETH Zurich) and catchment groundwater (University of Zurich) data available on request. WSL also operates EnviDat (718 datasets, DOI-referenced, CKAN API, harvested into opendata.swiss). WSL is a NAWA FRACHT partner alongside BAFU and Eawag.

Existenz.ch (community project, unofficial)

The de-facto developer API for FOEN hydrology; Aare.guru is built on it. JSON-only endpoints wrapping FOEN hydrology and MeteoSwiss SwissMetNet at 10-minute cadence, CORS *, Datasette instance with CSV export and station/parameter tables browsable at api-datasette.konzept.space, and an InfluxDB 2 long-term archive queried with Flux. Free for public/non-commercial use; BAFU data requires credit and link. Source: api.existenz.ch

Primary sources: bafu.admin.ch · hydrodaten.admin.ch · opendata.swiss · geocat.ch · eawag.ch · wsl.ch · envidat.ch · datalakes-eawag.ch · api.existenz.ch · data.bs.ch · ag.ch · zh.ch · ge.ch · ti.ch · vd.ch · data.tg.ch · uwe.lu.ch · bvd.be.ch · meteoswiss.ch · swisstopo.ch · trinkwasser.ch · data.stadt-zuerich.ch